

Jialuo (Eric) Chen

📍 85 Wood St., Toronto 📞 365-996-2310 ✉ jialuo.chen@mail.utoronto.ca [in linkedin](#) [github](#) [website](#)

Education

University of Toronto

September 2022 – June 2027

Bachelor of Applied Science and Industrial Engineering with PEY-COOP, focus in AI & ML

Toronto, ON

- **Minor:** Computer Science, Artificial Intelligence in Engineering, Engineering Business | **Annual GPA: 3.65**
- **Relevant Courses:** Bayesian Optimization & Sequential Decision Making (MIE498 Thesis), Operations Research (MIE262, MIE263, MIE365), Machine Learning (MIE370), Deep Learning (APS360), Artificial Intelligence (MIE369), RL (MIE369), Database Systems (MIE353), Data Structures & Algorithms (MIE245), Data Science (MIE223), Statistics (MIE237)

Publications

HMITL: Manager-Governed LLM Iteration with Guardrails and Rollback for

IEEE ICHI 2026

Reproducible Healthcare Machine Learning Pipelines

- Jialuo Chen, Haijing Wang, Siyu Shao

Accepted, Data Challenge Workshop

Role-Aware Learning for Bayesian Optimization under Uncertainty

NeurIPS 2026

- Jialuo Chen, Justin Beland

Manuscript under submission

Research Experience

Data-Driven Decision Making Lab, University of Toronto

May 2026 – Present

Machine Learning Research Intern — Advisor: Prof. Scott Sanner

Toronto, ON

- Designed a bounded **AR/XR world-model** architecture for physical-task assistance, representing state as a scene graph G_t plus temporal memory M_t rather than a generic AR chatbot layer.
- Mapped **Quest 3**, OpenXR/Meta SDK, scene anchors, segmentation outputs, and object labels into a perception-to-memory pipeline for objects, states, affordances, and task progress.
- Prepared research-facing architecture notes comparing SLAM/reconstruction-dependent and scene-API-first designs, with emphasis on persistent anchors, testable interfaces, and task-grounded assistance.

Li Shuai Research Group, Shanghai Jiao Tong University

January 2026 – Present

Remote Machine Learning Research Assistant — Advisor: Prof. Shuai Li

Shanghai, China

- Developing two parallel research threads in reasoning-oriented sequence modeling: **reasoning-basin dynamics** and **linear attention** / **LSA** for long-context inference.
- Focused primarily on the LSA direction, building controlled prototypes and diagnostics for attention efficiency, context retention, and reasoning stability without exposing method-specific unpublished details.
- Maintained literature notes and evaluation plans that separate implementation gains from publishable algorithmic claims, supporting cleaner comparison against existing efficient-attention and reasoning methods.

JB Research Group, University of Toronto

August 2025 – May 2026

Machine Learning Research Assistant — Advisor: Prof. Justin Beland

Toronto, ON

- Formulated **Role-Aware Bayesian Optimization under Uncertainty (RA-BOUU)**, a robust BO framework that updates a fixed Bayes-risk posterior through exact-control, executed-input, and aleatory coordinate roles.
- Implemented **executed-block kernels** including Positive Exponential Mixture (PEM) and Rational Quadratic (RQ) factors, preserving the inherited BOUU posterior algebra while changing how realized observations transfer covariance to $J(x)$.
- Developed a decision-aligned aleatory acquisition rule that selects environmental nominees by expected one-step uncertainty reduction in the robust target rather than pointwise simulator variance.
- Ran matched synthetic and **Safe-Control-Gym Quadrotor2D** experiments with shared initialization, robust re-evaluation, external baselines, and ablations; wrote method, theory, and experiment sections for a NeurIPS 2026 submission.

Industry Experience

JANA Corporation

May 2025 – May 2026

Junior Software Developer

Aurora, ON

- Translated research documentation on probabilistic risk modeling into production-oriented **Python** modules for likelihood-of-failure analysis in large-load metering systems.
- Converted historical incident records into quantitative risk features and diagnostic outputs used to compare threat influence and reliability evolution over time.
- Delivered production-facing changes in **React**, **TypeScript**, and **C#** for an enterprise pipeline-risk web application, improving model configuration, data filtering, and UI reliability.

Selected Projects

NeuroCommute | *Personalized Routing, A*, LLM Agent, GTFS/OSM*

December 2025 – Present

- Built a Toronto multimodal routing prototype over TTC/GO/UP transit data and OSM-derived graphs, supporting transfer-aware **A*** search across time, walking, transfers, fare, congestion, and intersection penalties.
- Implemented an optional **LLM preference layer** that maps natural-language commute goals into bounded routing weights, keeping the graph planner deterministic, auditable, and separable from language-model explanations.
- Generated diverse route candidates through scenario portfolios such as fastest, least-walking, fewest-transfers, cheapest, and safer routes, then reranked outputs based on user-specific constraints.
- Used early validation from 89 survey responses to prioritize the downtown-student beachhead and safer, lower-friction commuting scenarios for the Toronto beta.

LineToLive Sketch-to-Face GAN [link](#) | *Computer Vision, GAN, PyTorch*

June 2024 – August 2024

- Studied feature entanglement in standard cGAN pipelines such as Pix2Pix, focusing on why line-to-face reconstruction often loses high-frequency facial detail.
- Built a three-stage generator pipeline with **U-Net** and **PatchGAN** components to separate facial structure, colorization, and texture refinement on the CUHK sketch-face dataset.
- Reduced L1 reconstruction loss by **25.1%** against a monolithic baseline while producing clearer facial geometry and more stable perceptual outputs.

ADHD Scholarship Copilot [link](#) | *Human-AI Interaction, RAG, ChromaDB*

November 2025 – Present

- Designed a **Cognitive Support System** to reduce executive dysfunction for ADHD users, utilizing **Chain-of-Thought (CoT)** prompting with Claude AI to decompose complex scholarship application into atomic, low-friction micro-tasks.
- Implemented a **RAG architecture** using ChromaDB to parse the scholarship's elements against saved user profiles, ensuring privacy-preserving, context-aware retrieval to establish an accurate, efficient application cycle.
- Developed a **cognitive enhancement interface** ("Focus Shield") based on Flow Theory, integrating visual occlusion rules to extend user attention spans during high-cognitive-load tasks.

Skills

Programming: Python, C, Java, SQL, JavaScript/TypeScript, C#, Shell Scripting, Linux, R, Matlab

AI/ML/Data: Bayesian Optimization, Bayesian Inference, Machine Learning, Deep Learning, NLP (Transformers, Linear Attention), Reinforcement Learning (MCTS, Q-Learning), GenAI (RAG, Agents), CUDA

Framework/Tools: Git, PyTorch, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PostgreSQL, Docker, JupyterLab

Additional Skills: Communication, Teamwork, Agile Development, Problem-Solving, Collaboration, Leadership, Jira